

Deep Learning-Based Classification of Lung and Colon Cancer Subtypes Using Histology Images

Vishnu Kant

Chitkara University Institute of Engineering and Technology,
Chitkara University, Rajpura, India
vishnu.kant@chitkara.edu.in

Bura Vijay Kumar

School of Computer Science and Artificial Intelligence, SR
University, Warangal - 506371, Telangana, India
vijaykumar.bura@gmail.com

Abstract— Among the main causes of cancer-related death worldwide are lung and colon cancers. Early malignant tumour discovery and precise classification of them define both appropriate treatment and better patient outcomes. This paper provides a convolutional neural network (CNN) model aimed in classifying lung and colon cancer subtypes including colon benign tissue, lung adenocarcinoma, lung benign tissue, and lung squamous cell carcinoma. With 500 images in every class, the model was developed on a 2,500-histology image Kaggle dataset. Since every class gains almost perfect values in accuracy, recall, and F1 scores, our results show a remarkable 99% classification accuracy. To give perfect separation between benign and malignant tissues, the proposed CNN design successfully finds intricate patterns in the photos. The study exceeds conventional diagnostic methods since it reveals the great promise of CNNs as a fast, highly accurate non-invasive tool for cancer diagnosis.

Keywords— Artificial Intelligence, Deep Learning, Lung Colon Cancer Classification, Model Training, Proposed CNN Model.

I. INTRODUCTION

Lung colon cancer illnesses are among the most often occurring and fatal forms of cancer known to exist worldwide. Still the primary cause of cancer-related mortality is lung cancer, which can show up as squamous cell carcinoma and adenocarcinoma. Colon cancer, including adenocarcinoma and other types, causes similar worldwide burden on health. These illnesses are difficult, with their aggressive character and many histological subtypes that demand precise and early diagnosis for suitable therapy. Early detection helps to increase patient survival rates by means of customized therapy regimens and early intervention. Nevertheless, given their invasiveness, expense, and generally poor accuracy, traditional diagnostic techniques including imaging and biopsies typically fail. The proposed convolutional neural network model addresses these issues by providing a rapid, non-invasive, very accurate technique of cancer diagnosis. Usually overlooked by traditional methods, the CNN architecture has been modified to detect intricate trends in histology images. The CNN model shines in pattern recognition and feature extraction, hence maybe our approach to cancer detection should be changed. This work emphasizes the probable application of the suggested CNN model in clinical settings as it seems to prove its efficiency in exactly

classifying several lung and colon cancer subtypes. By means of deep learning, we wish to support the ongoing work in cancer detection, so improving patient outcomes using this capacity. The results of this work emphasize the importance of adding innovative computational techniques into medical diagnosis, therefore boosting the battle against cancer.

II. LITERATURE

The most recent advances in deep learning essentially the convolutional neural network have enhanced the advanced field of lung colon cancer classification and targets to attain the high accuracy levels, essential for the medical imaging of lung colon cancer. Many of the transformation techniques applied in the framework eliminate properties and provide a better data representation. Models for handling histological pictures are Shufflenet, mobilenet, and squeezenet. Then, principal component analysis helps to reduce deep features obtained from these models. Four different machine learning methods are given the final lowered characteristics, therefore attaining a top accuracy of 99.6%. Using lightweight deep learning models, the proposed approach clearly differentiates lung from colon cancer types with less features and reduced computational cost [1]. The work proposes modifications to provide pre-trained CNN models using improved augmentation techniques to identify lung and colon cancer using histomorphology images. Using accuracy, F1-score, precision, and recall [2], performance was assessed for all models generating very remarkable results spanning from 96% to 100% accuracy. Researchers find that convolutional neural networks pretty successfully categorize histological images encompassing numerous cancer kinds, including brain, skin, breast, lung, and colorectal cancers. This work employs VGG-19 for feature extraction and ResNet 50 for classification of lung cancer images [3]. To get best detection performance, the new architecture employs around 0.65 million trainable parameters and 6.4 million flops. This design shows modern performance unlike that of previous methods [4]. Under this approach, a pre-trained network generates features from the dataset while test images are categorized using a multi-class classification technique. All models exhibited exceptional classification performance when achieving accuracy between 97.32% and 99.79% for lung-colon cancer picture classification [5]. This novel combination model seeks to increase model robustness, lower overfitting, and enhance feature extraction. With perfect precision and memory for identifying certain malignant and non-cancerous tissue, the results expose an amazing 99.44%

classification accuracy, much above traditional approaches [6]. Recent advances in deep learning have drawn interest on its application to medical imaging analysis, particularly in the diagnosis of colon cancer and lung cancer using histomorphology images. This study aims to improve present pre-trained CNN models like Xception, DenseNet 201, ResNet101, InceptionV3, by means of enhanced augmentation strategies, so enhancing classification [7]. After the high-performance filter enhanced the best-performing algorithms, weighted averaging integrated them under an ensemble learning system. With accuracies of 99.85% for two-class, 98.70% for three-class, and 98.96% for five-class datasets, the proposed ensemble learning model proves to be far better than present approaches in classification accuracy across all datasets [8]. Based on the squared change in network loss, this paper suggests a new selection criterion termed architecture saliency for optimum NAS architectures. The paper ends with demonstrating via experiments employing an ensemble of ResNet50 and Elastic Net the efficiency of the parametric method in pixel and image-level classification in detecting lung and colon cancer cells [9]. Generation of a collection of random individuals happens first then mutation, recombination, and selection. Every generation the lowest loss values are tossed out and replaced with more competitive members statistically benchmarked against the new recommended strategy [10]. Recent studies on cancer type classification and improvement of diagnostic accuracy have revealed amazing performance of deep learning approaches [11]. After 12 Adams optimizer training cycles using a 128-based batch size, we analyzed the produced accuracy and loss graphs in great detail. The findings of the study show that the EfficientNet B7 model is well-suited for this classification challenge since it is obviously and with high recall categorized different kinds of lung and colon cancer [12].

III. INPUT DATASET

The dataset depicted in Fig. 1 was initially obtained from the open-source Kaggle platform and contains images of lung and colon carcinoma. Five distinct sets of 500 photos are comprised of colon adenocarcinoma, colon benign tissue, lung adenocarcinoma, lung benign tissue, and lung squamous cell cancer. By means of data augmentation used to alter the dataset, preprocessing the pictures helps to Overheads of data augmentation thus boosting the generalizing ability of the model encompassing numerous image orientations and scenarios where rotation, flipping, and scaling. Helping this is standardizing the pixel values and reducing them to fitable CNN input layer size. This diverse and balanced dataset provides a robust platform for training and assessment of the proposed CNN model since it assures whole coverage of the relevant cancer subtypes and thus helps to construct an accurate and reliable classification system.

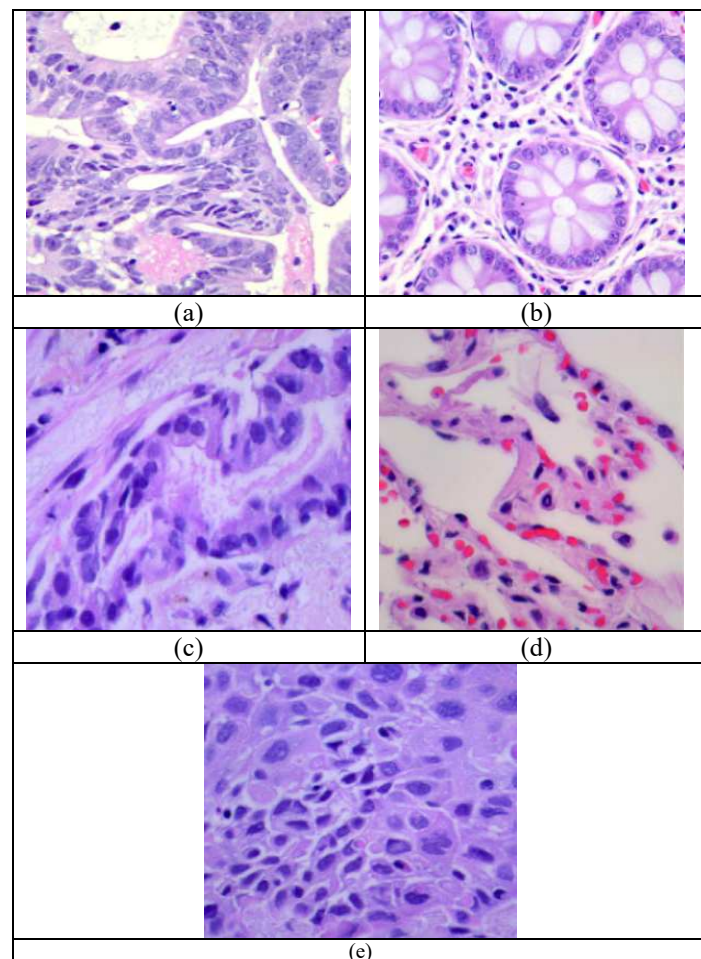


Fig. 1 Dataset image for (a) Colon Adenocarcinoma (b) Colon Benign Tissue (c) Lung Adenocarcinoma (d) Lung Benign Tissue and (e) Lung Squamous Cell Carcinoma

IV. PROPOSED CNN MODEL

Seeking to effectively classify lung and colon cancer subtypes, sought to use the vast feature extraction capability of deep learning. The proposed CNN model architecture as shown in Fig. 2 Several convolutional layers in the model progressively capture complex patterns in histopathology images, including structural changes unique to every disease type and texture and following every convolutional layer is a max-pooling layer that lowers spatial dimensions and computing effort while yet maintaining the most relevant information. ReLU and other activation methods introduce non-linearity, which enables the model to grasp complex connections among pixel values. Comprising completely connected dense layers aggregating the obtained data into high-level representations, the last layers pass via a softmax layer for categorization into the five target classes. Extensive hyperparameter tuning—that is, learning rate and batch size—allows this architecture to be changed to reach robust performance and great accuracy.

V. PROPOSED METHODOLOGY

This paper offers the four-phase method to develop and evaluate a suggested CNN model for lung colon cancer disease categorization shown in Fig. 2. Four procedures define it: dataset gathering, dataset preprocessing, model training and validation and model assessment and performance matrix.

Phase 1: Dataset Collection

First comes compiling photos for the open-source Kaggle platform using extensive and high-quality content. Every class comprised of 500 photographs; the dataset was evenly divided among the five separate classes: colon cancer; benign tissue from colon; lung adenocarcinoma; benign tissue from lung squamous cell after analysis of the total images of 2500 images. This balanced dataset assures that the model is trained on a wide range of samples, therefore catching the variations between benign and malignant tissues as well as within every cancer subtype. The images were selected to depict many phases and symptoms of the illnesses, therefore providing a sufficient base for the upcoming stages of model building and validation.

Phase 2: Data Preprocessing

Through preprocessing the obtained images, they aligned with CNN model input. To meet input layer requirements, the images were resized to a standard resolution suited for CNN architecture. Moreover, pixel normalisation was applied to standardise the image data thus guarantee continuous numerical ranges, so enhancing the model's training efficiency. More generally, as a kind of data augmentation, rotating, flipping, and scaling increased the generalizability of the idea yet more. These techniques artificially improve the variety of the training dataset and therefore reduce the danger of overfitting by mimicking changes in image orientations and conditions, so improving the performance of the model on unseen data.

Phase 3: Model Training

With an eye toward building the proposed CNN model, specifically designed to classify the cancer subtypes depending on processed images, the third phase centers on CNN's architecture consisting on many convolutional layers that extract features from the images, so collecting important patterns like texture and form suggestive of particular cancer subtypes. Dropout layers created inside the network randomly deactivate a percentage of neurons during training, hence lowering overfit. The last layers are completely connected dense layers that aggregate the recovered properties into high-level representations, therefore generating the output layer with a softmax function categorizing the input pictures into the five target classes.

Phase 4: Model Evaluation and Validation

Here we measure CNN model performance on the last phase evaluation as well as on training. Under the direction of the loss function led by the Adam optimizer, the training step

consists in feeding the pre-processed images into the model. Confusion matrices were also utilized to assess the performance of the model in order to identify areas maybe needing development. Part of the training over numerous epochs was early stopping utilized to stop overfitting following the stop improvement of the validation loss. Since they confirmed the CNN model's accuracy in spotting lung and colon cancer subtypes, the results of this phase underlined the possibility for pragmatic implementation of the CNN model in cancer diagnosis.

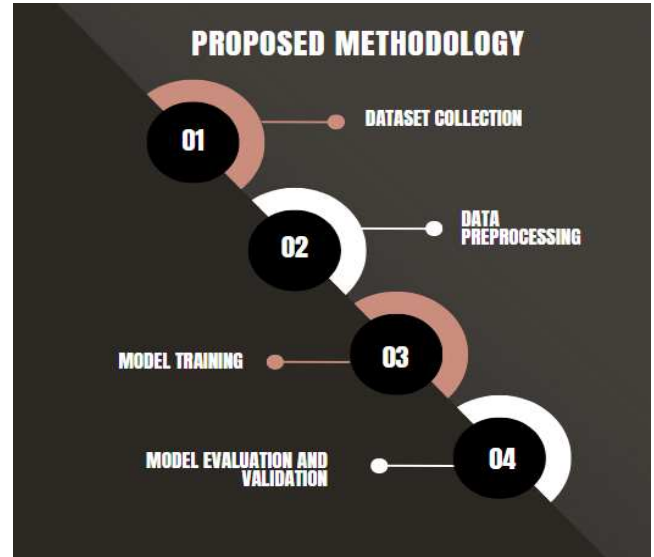


Fig. 2 Proposed Methodology

VI. RESULTS

Effective learning allowed the proposed CNN model to demonstrate over 99% peak validation accuracy. The graphs on loss and accuracy exhibit decent convergence. Apart from the steady validation accuracy, the continuous decline in training and validation losses highlights the model's ability to generalize efficiently among the five classes of lung colon cancer illness.

A. Classification Report Analysis

Based on exact performance overall cancer types, the Table 1 classification report demonstrates a general accuracy of 99%). Every class displays accuracy, recall, and F1-scores near to or equal to 0.99, therefore verifying the precision of the model in class division. Especially getting 100% recall, the colon benign tissue class and colon adenocarcinoma reveal that the model precisely recognizes all real events of both subtypes. Moreover, perfect accuracy and recall displayed by lung benign tissue confirm the model's effectiveness in managing several histological characteristics. Still, these fall in a somewhat reasonable range and F1-scores suggest strong overall categorization. This outstanding degree of performance on all criteria emphasizes the strength of the suggested CNN model and its likely use in clinical diagnostics for colon and lung cancer diagnosis.

Class	Precision	Recall	F1-Score	Support
Colon Adenocarcinoma	0.99	1.00	0.99	500
Colon Benign Tissue	1.00	0.99	0.99	500
Lung Adenocarcinoma	0.97	0.97	0.97	500
Lung Benign Tissue	1.00	1.00	1.00	500
Lung Squamous Cell Carcinoma	0.98	0.97	0.97	500
Accuracy			0.99	2500
Macro Avg	0.99	0.99	0.99	2500
Weighted Avg	0.99	0.99	0.99	2500

Table 1. Classification Report Analysis

B. Training and Validation loss Analysis

The fast beginning drop in both training and validation loss as seen in Fig. 3 implies to the model quick identification of the basic patterns required for classification from the loss plot. The declining loss numbers in epoch five expose CNN's effectiveness in mistake reduction. The loss lowers much further as the epochs pass and settles around epoch 10 when training and validation curves differ very little. This thorough study of the two curves indicates a well-tuned model that lowers overfit by means of a solid balance between bias and variance. Epoch 19 produces the best outcome thus verifying that the selected termination point is ideal and guaranteeing the model maintains its predicted accuracy without too much training. This stability implies that the model has effectively converged depending on the close alignment of the training and validation loss curves, therefore avoiding overfitting or underfitting situations.



Fig. 3 Training and Validation loss Analysis

C. Training and Validation Accuracy Analysis

Reflecting the quick fit of the model to the dataset, the accuracy curve as shown in Fig. 4 obviously shows a spike in both training and validation accuracy within the first few epochs. By epoch five, both accuracies exceed 85%; they keep increasing steadily until they level off at 98–99% accuracy levels. On both seen and unprocessed data, the very generalizable model proposed by the closeness of the training

and validation accuracy curves performs well. The little variations seen toward the latter epochs within tolerable boundaries show that the model preserves strength free from substantial overfitting. For the classification task, the accuracy research reveals generally that the model's design and training strategy are rather successful. The little differences in validation accuracy lead to great generalizability and show that the model maintains resilience in handling unexpected inputs.

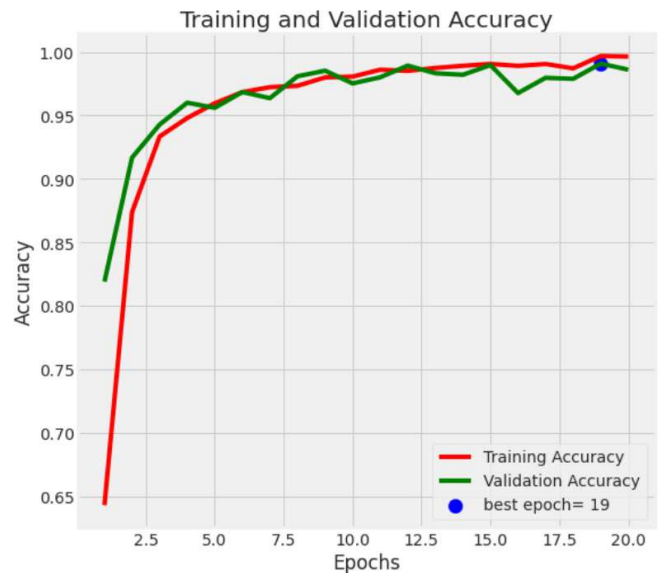


Fig. 4 Training and Validation Accuracy Analysis

D. Confusion Matrix Analysis

The confusion matrix presented in Fig. 5 shows that the suggested CNN model very well detects several forms of cancer. Presenting real positives, the quite high diagonal values show the model's ability for exact class prediction. When adenocarcinoma types are mixed with their benign counterparts in a few cases, non-diagonal entries suggest misclassifications—rare occurrence. This suggests a required potential for therapeutic uses as the model obviously separates benign from malignant tissues. The strong sensitivity and specificity rates for every class show that the model not only points out the correct events but also reduces false positives and negatives, therefore helping to reduce diagnosis errors. All things considered, the confusion matrix supports the conclusion that the CNN model offers a reliable and suitable tool for lung and colon cancer classification, hence delivering promised greater accuracy in actual diagnosis settings.

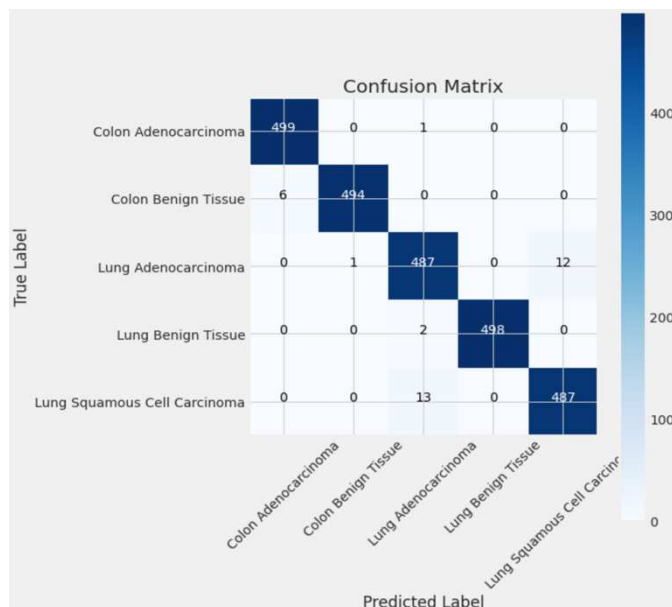


Fig. 5 Confusion Matrix Analysis

VII. CONCLUSION

Confusion matrix analysis and performance assessments point to the suggested CNN model for lung colon cancer displaying a strong performance with appropriate accuracy. The model obtained an overall accuracy of 99% with remarkable accuracy, recall, and F1-scores across all five classes; colon adenocarcinoma, colon benign tissue, lung adenocarcinoma, lung benign tissue, and lung squamous cell cancer. The results highlight and define suitable clinical diagnosis and treatment planning as the model can accurately differentiate benign from malignant tissues. Using deep learning to address the drawbacks in current diagnosis methods, the proposed CNN model offers a non-invasive, fairly reasonable cost strategy for cancer detection. The method clarifies the medical diagnostic supporting the proper and customized healthcare treatments for lung colon cancer disease. Under real-world clinical settings, future research will concentrate on improving the model, increasing the dataset, and verifying its usefulness in daily medical practice.

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